PulseIQ Phase 1 — Submission Report

Workforce and operational analytics · Source cut-off: 30 September 2026

1. Project overview

This package is based on the supplied fictional Zenith Nova Technologies / Aura Workforce workbook. The original workbook and CSV exports are retained.

2. Source inventory

Workbook sheets exported: 30; data dictionary rows: 371; data contracts rows: 42; metric definitions: 28; DQ catalog checks: 14.

3. Included components

- ETL pipeline with exact-duplicate quarantine, audit metadata, and table profiling.
- PostgreSQL canonical dimension/fact schema and illustrative analytical views.
- Automated baseline data-quality checks with PASS / FAIL / REVIEW statuses.
- Streamlit dashboard with source-table explorer, headcount chart when fields are identifiable, metric dictionary, and DQ catalog.
- Python requirements, Dockerfile, smoke tests, README, and source inventory.

4. Validation caveats

The original starter ZIP, validation vault, and official report template were not supplied. The SQL schema requires mapping to the actual CSV headers. Metric formulas and all 14 DQ rules require rule-by-rule reconciliation before claiming full validation. No screenshots, run logs, or passing test results have been fabricated.

5. Run instructions

Install requirements with pip install -r requirements.txt, then run python src/etl_pipeline.py, python src/quality_checks.py, streamlit run app.py, and pytest -q from the repository root.

6. Output interpretation

A REVIEW status means the implementation could not safely identify the required field from the source headers; it is not a passing result. Review outputs/dq_results.csv and the report before submission.